Meowximizer

Short manual . free and open source (AGPL-3.0-or-later) . github.com/SpeakingColors

Meowximizer is a sonic enhancer, bass enhancer and exciter in one. It splits the sound into three phase-aligned bands and works on each: bass contour and harmonics, mid and high timing and dynamic lift, an air exciter, a squeeze section (compression, push and opto-style smoothing with several styles), warm saturation and an output limiter. Aim: put it on a lead or bass and it gets thicker, smoother and louder without losing punch.

Quick start

• Pick a preset close to your source (Thick Supersaw, Dance Master Lead, Mix Bus Glue, Kick Beef).
• Choose a Squeeze style in the menu: it sets the Squeeze, Push and Opto knobs and changes how the chain works.
• Adjust Tone (bass, process, air) by ear, then Warm for saturation.
• Set Ceiling for the limiter and match Output to the bypassed level.
• The round gear button in the header opens Settings: the plugin version, a link to the GitHub page and the releases page (it opens your browser; the plugin itself never goes online), and a switch for the info bubbles. Hover any control for about one second to see what it does; the round i button also turns the bubbles on or off.

How the window works

• Each knob sits directly under the display it changes; the displays react to the knobs and the knobs react to the displays.
• Meters show the real signal in dB with a dB scale: input and output on the far right.
• Hover a control for about one second to see an info bubble. Turn bubbles off with the round i button in the header.
• The Limiter switch in the header adds a safety limiter at the output (useful while experimenting).
• Presets are in the header; the dice picks random settings (some controls are excluded on purpose).

Controls

Tone

Air exciter	Choice: Shipped: tanh(5x) / PurrOne stack (gritty when pushed) (default Shipped: tanh(5x))	The shaper that makes the Air harmonics. The stack version gets gritty and fuzzy when Air is pushed high.
Lo contour	0 to 10 default 3	Low contour: a gentle low shelf for weight and warmth in the bottom end.
Bass harmonics	0 to 100 % default 25 %	Bass harmonics: adds overtones to the bass so it stays audible on small speakers.
Process	0 to 10 default 5	Process: the sonic-maximizer effect. Delays the lows and mids a tiny bit behind the highs and lifts the highs when the mids are quiet, for clarity and punch.
Air	0 to 100 % default 25 %	Air: a smooth high shelf at 9 kHz (up to +4.5 dB) for sheen, plus an exciter that adds new harmonics above 5 kHz for brilliance and excitement. The exciter runs 2x oversampled, so it stays smooth.

Squeeze

Squeeze mode	Choice: Gentle, OTT style, Dance Master, Glue, Slow Smooth, Punchy, Tight Lock, Lift, Hardstyle Lock, Highs Tamer, Thick Supersaw (default Gentle)	Squeeze style: each style is a different chain. Gentle keeps dynamics natural. OTT presses the loud down and lifts the quiet up hard. Dance Master is a controlled multiband compressor with soft knee, auto release and a limiter on each band. Glue is mild and sticky. Slow Smooth is slow and smooth, like an opto. Punchy lets the attack through. Tight Lock clamps hard and fast. Lift brings up quiet parts. Hardstyle Lock is fast and firm with extra low-end control. Highs Tamer holds back harsh highs on leads (good after a reverb). Thick Supersaw is a slow, soft squeeze that pairs with the Opto knob to fatten supersaw leads.
Loudness match	Choice: Auto, Off, On (default Auto)	Loudness match: a slow level follower lifts the processed sound back up to the level that went in, so heavy squeeze or opto settings never lose volume or loudness. Auto switches it on as the Opto amount grows. It only ever lifts, and never past the Ceiling.
Squeeze	0 to 100 % default 30 %	Squeeze: how much each of the three bands is evened out, pressing loud parts down and lifting quiet ones up. More = denser.
Push	0 to 18 dB default 0 dB	Push: raises the level going into the squeeze and the limiter for more loudness.
Opto	0 to 100 % default 0 %	Opto: extra LA-2A style leveller on top of whatever the squeeze style already uses. Slow, smooth, program-dependent compression that thickens leads and bass. 0 = only what the style brings.

Saturation and ceiling

Warm character	Choice: Shipped: tanh, Drive (opamp tanh), Transistor (even harmonics), Fuzz (lopsided), Stack: all three, grows with Warm (default Shipped: tanh)	What the Warm knob does. Shipped is the plain tanh. The others are the PurrOne colour stages, oversampled 2x. Stack grows from drive to transistor to fuzz as you push Warm.
Clipper position	Choice: Pre: on the input / Post: just before the limiter (default Post: just before the limiter)	Where the Clip stage sits. Post (before the limiter) keeps the squeeze natural and is what mastering clippers do. Pre clips the input before everything.
Warm	0 to 100 % default 20 %	Warm: saturation on the whole signal. Rounds peaks and adds body. With a Warm character other than tanh it adds disto, transistor or fuzz grit when pushed.
Clip	0 to 100 % default 0 %	Light peak clipper: pushes the signal up to 9 dB harder into a smooth clipper at the Ceiling, then the limiter. Shaves the peaks so the whole thing can sit much louder.
Ceiling	-12 to 0 dB default -0.3 dB	Ceiling: the output limiter never lets the signal go above this level.

Level

Mix	0 to 100 % default 100 %	Mix: dry/wet blend for parallel processing.
Output	-12 to 12 dB default 0 dB	Output: final level. Match it with the bypassed level when comparing.

Recommendations

• Leads: Thick Supersaw or Dance Master styles, Opto 50 to 85, then lower Mix if it sounds crowded.
• Bass: Tight Lock or Smooth styles, keep Bass harmonics moderate, check mono and small speakers.
• Mix bus or master: Glue or Gentle style, low Squeeze, Ceiling at -1 dB.
• Hardstyle kicks: Hardstyle Lock style with Push 1 and a Ceiling that leaves room for your mastering chain.

Presets

Init, Loud Master, Warm Glue, OTT Hardcore, Vocal Air, Bass Boost, Drum Smack, Thick and Smooth, LA-2A Lead Thicken, LA-2A Bass Smooth, Dance Master Lead, Dance Master Bass, Reverb Lead Control, Goodizer Light, Goodizer Medium, Sub Bass Thick, Pluck Punch, Pad Smooth, Vocal Leveller, Hardstyle Kick Glue, Supersaw Control, Mix Bus Glue, Glue Bus, Smooth Lead (Slow), Punchy Drums, Tight Bass Lock, Quiet Lift, Hardstyle Lead Lock, Harsh Highs Tamer, Thick Supersaw, Kick Beef (Parallel), Mastering Touch

Interface tour

Pink numbers mark controls, teal letters mark displays. Every number is explained in the tables on the next pages. The picture is a screenshot of the real compiled plugin window.



Signal flow

Input (+ Push) > Tone shaping (Lo contour, Bass harmonics, Process, Air) > Squeeze (three band, style, Opto) > Warm > Clip > Ceiling limiter > Mix and Output. Loudness match keeps the level of heavy styles.

Header bar

1	**Preset menu**	Factory presets (our own). Use the arrows to step through them.
2	**Randomize**	Set the controls to random values (output level and on/off switches are kept).
3	**Nasty range**	Nasty range: lets Drive, Fuzz, Squeeze, Warm and the other heavy knobs go far past 100 % for really nasty, OTT and Vengeance style sounds. Off keeps every knob at its normal top.
4	**Limiter**	Safety limiter on the output: keeps the level from ever passing 0 dBFS, smooth and transparent. Turn it off if you want the raw output (for example to let your own limiter do the work).
5	**Settings**	Settings: version, link to the GitHub page for new versions
6	**Info bubbles (i)**	Info bubbles: hover a control for one second to see what it does. Click to turn them off or on.

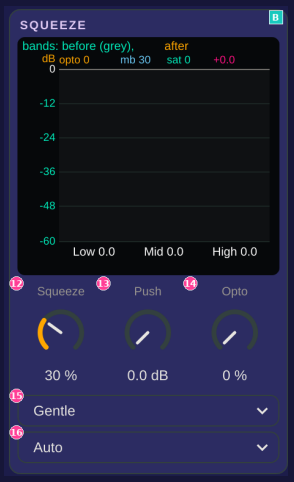
A. Tone



Display A: Tone curve, the sum of the four tone controls. Horizontal: frequency (log), vertical: dB. White line: the total. Coloured fills: Lo contour (blue), Bass harmonics (orange), Process (pink), Air (teal), with their live dB values on top. The vertical scale follows the knobs from -6 to +12 dB.

7	**Lo contour**	Low contour: a gentle low shelf for weight and warmth in the bottom end. (shown: 3.0)
8	**Bass harmonics**	Bass harmonics: adds overtones to the bass so it stays audible on small speakers. (shown: 25 %)
9	**Process**	Process: the sonic-maximizer effect. Delays the lows and mids a tiny bit behind the highs and lifts the highs when the mids are quiet, for clarity and punch. (shown: 5.0)
10	**Air**	Air: a smooth high shelf at 9 kHz (up to +4.5 dB) for sheen, plus an exciter that adds new harmonics above 5 kHz for brilliance and excitement. The exciter runs 2x oversampled, so it stays smooth. (shown: 25 %)
11	**Air exciter**	The shaper that makes the Air harmonics. The stack version gets gritty and fuzzy when Air is pushed high. (shown: Shipped: tanh(5x))

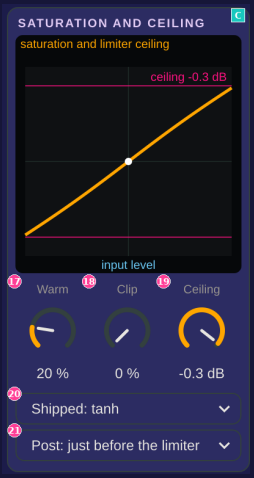
B. Squeeze



Display B: The three squeeze bands (Low, Mid, High). Grey bars: level before the squeeze; coloured bars: level after, with the value in dB under each band. The label top right says whether loudness match is on.

12	**Squeeze**	Squeeze: how much each of the three bands is evened out, pressing loud parts down and lifting quiet ones up. More = denser. (shown: 30 %)
13	**Push**	Push: raises the level going into the squeeze and the limiter for more loudness. (shown: 0.0 dB)
14	**Opto**	Opto: extra LA-2A style leveller on top of whatever the squeeze style already uses. Slow, smooth, program-dependent compression that thickens leads and bass. 0 = only what the style brings. (shown: 0 %)
15	**Squeeze mode**	Squeeze style: each style is a different chain. Gentle keeps dynamics natural. OTT presses the loud down and lifts the quiet up hard. Dance Master is a controlled multiband compressor with soft knee, auto release and a limiter on each band. Glue is mild and sticky. Slow Smooth is slow and smooth, like an opto. Punchy lets the attack through. Tight Lock clamps hard and fast. Lift brings up quiet parts. Hardstyle Lock is fast and firm with extra low-end control. Highs Tamer holds back harsh highs on leads (good after a reverb). Thick Supersaw is a slow, soft squeeze that pairs with the Opto knob to fatten supersaw leads. (shown: Gentle)
16	**Loudness match**	Loudness match: a slow level follower lifts the processed sound back up to the level that went in, so heavy squeeze or opto settings never lose volume or loudness. Auto switches it on as the Opto amount grows. It only ever lifts, and never past the Ceiling. (shown: Auto)

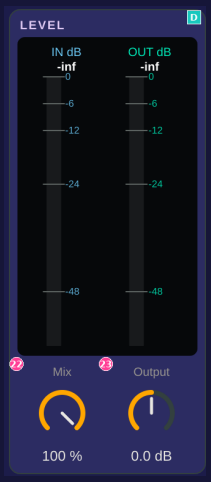
C. Saturation and ceiling



Display C: Transfer curve of Warm plus Clip and the Ceiling. Horizontal: input level, vertical: output. The orange line shows the combined shape, the pink line is the ceiling, the white dot is the live level.

17	**Warm**	Warm: saturation on the whole signal. Rounds peaks and adds body. With a Warm character other than tanh it adds disto, transistor or fuzz grit when pushed. (shown: 20 %)
18	**Clip**	Light peak clipper: pushes the signal up to 9 dB harder into a smooth clipper at the Ceiling, then the limiter. Shaves the peaks so the whole thing can sit much louder. (shown: 0 %)
19	**Ceiling**	Ceiling: the output limiter never lets the signal go above this level. (shown: -0.3 dB)
20	**Warm character**	What the Warm knob does. Shipped is the plain tanh. The others are the PurrOne colour stages, oversampled 2x. Stack grows from drive to transistor to fuzz as you push Warm. (shown: Shipped: tanh)
21	**Clipper position**	Where the Clip stage sits. Post (before the limiter) keeps the squeeze natural and is what mastering clippers do. Pre clips the input before everything. (shown: Post: just before the limiter)

D. Level



Display D: Input and output meters in dB with the dry/wet bar (pink dry, teal wet).

22	**Mix**	Mix: dry/wet blend for parallel processing. (shown: 100 %)
23	**Output**	Output: final level. Match it with the bypassed level when comparing. (shown: 0.0 dB)

Under the hood

• Squeeze works on three bands (low, mid, high) and picks the style chain: choosing a style moves Squeeze, Push and Opto to match it. Extended ranges: Squeeze to 250 % (OTT territory), Push to 36 dB.
• Loudness match is a slow closed-loop follower that lifts the squeezed sound back to the level that went in. Auto turns it on for OTT Nasty and Vengeance Slam, off for the other styles.
• Opto is an LA-2A style leveller added on top: slow and program dependent.
• Warm characters other than tanh use the PurrOne colour stages, oversampled 2x. Stack grows from drive to transistor to fuzz as Warm is pushed.
• Clip: a smooth clipper at the Ceiling, up to 9 dB harder, then the limiter. Post (before the limiter) is the mastering way; Pre clips the input.
• Tone curve is measured per component and scaled live by the knobs, so the display is the real response, not a drawing.

Install and use

• Windows: run the installer (.exe). macOS: open the installer (.pkg or .dmg). Linux: install the Flatpak with `flatpak install --user` and the .flatpak file, or copy the .vst3 folder to ~/.vst3.
• Everything works offline. Rescan plugins in your DAW, then look for the plugin under Studio 87.
• Report problems or ideas on the GitHub page of the plugin. Free to use, copy and change under the AGPL-3.0-or-later licence.